

Project ID : 25-26J-534

1. Topic (12 words max)

Digital Drive and Young Car Enthusiasts

2. Research group the project belongs to

TIM - Technology Integration and Management

3. Specialization of the project belongs to

Information System Engineering(ISE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

The consumer journey for vehicle purchasing has changed significantly. Nowadays, a search query or a quick glance through a social media feed sparks what used to be a physical showroom visit or a print magazine review. Social media platforms have turned into multi-purpose location, effectively functioning as a community hub, a 24/7 customer support desk and modern showroom. This shift demands that automotive brands adopt an elaborate, subjective technique to marketing in place of just promoting their products.

Connecting with younger consumers whose purchasing intentions are based on passion and community rather than just business is significant challenge for the automotive industry [1]. Recent trends show a noticeable change, with younger generations being more interested in owning classic, sports and collector cars than their predecessors [2]. The reason they actively participate in unique subcultures like classic car restoration is to spark interest, passion, and curiosity among the younger generation, including ladies, through sharing videos and pictures of such restored vehicles on social media [3] [9]. Unfortunately, social representations regarding classic car modification and restoration create unique marketing opportunities that remain highly untapped in Sri Lanka [4].

A significant research gap exists in how these automotive brands currently use social media platforms to interact with car enthusiasts [5] [8]. While the automotive industry is growing globally with the adaptation of digital marketing, limited research exists on the Sri Lankan context, specially focusing on young car

enthusiasts and restoration parties which perform poorly on platforms like Instagram, Facebook, TikTok and YouTube, are still failing to unlock its full potential through social representations [6] [7].

The research aims to address the key necessity to understand how social media can effectively engage young automotive enthusiasts in Sri Lanka's classic and sports car modification and restoration subculture, while increasing the customer base, to be a spark in classic, sports car modification and restoration in Sri Lanka, to boost enthusiasm and build a talented team as a brand internationally in Sri Lankan automotive industry for the restoration of vintage sports cars.

The study will offer useful frameworks for local workshops, by unlocking the digital transformation in emerging market automotive industry.

References

- [1] M. M. a. E. P. B. Cappellini, "The hidden work of coping: gender and the micro-politics of household consumption," *Journal of Marketing Management*, vol. 30, pp. 1597-1624, 2014.
- [2] Specialty Equipment Market Association (SEMA), "2023 Market Data Report: Young Enthusiast Vehicle Buyers," SEMA Research Reports, 2023.
- [3] S. D. a. P. M. R. Kumar, "Social Media Data Analytics in the Automotive Industry: Interactive Impact of Marketing Strategies," *International Journal of Information Management*, vol. 52, pp. 102-115, 2020.
- [4] M. M. E. P. a. P. D. B. Cappellini, "Driven by nostalgia: A study of classic cars' social representations among Italian youth," *Journal of Business Research*, vol. 88, pp. 416-423, 2018.
- [5] A. P. a. V. Mahajan, "Adoption of Social Media Marketing Strategies in Automotive Industry: A Developing Market Perspective," *Digital Business*, vol. 20, no. 2, pp. 100-115, 2022.
- [6] S. Devi, "Social Media as a Tool of Marketing: A Study of Indian Automobile Industry," *International Journal of Marketing Studies*, vol. 7, no. 2, pp. 111-120, 2015.
- [7] V. C. a. L. H. L. De Veirman, "Marketing through Instagram influencers: the impact of number of followers and product divergence on brand attitude," *International Journal of Advertising*, vol. 36, no. 5, pp. 798-828, 2017.
- [8] S. Xue, "Social Media Data Analytics in the Automotive Industry: A Study of the Interactive Impact of Marketing Strategies and User Ratings," *Advances in Economics, management and Political Sciences*, vol. 78, pp. 142-147, 2024.
- [9] Y. Han, "Self-Presentation of Youth Subcultures on Social-Media," *Communications in Humanities Research*, vol. 9, no. 1, pp. 20-26, 2023.

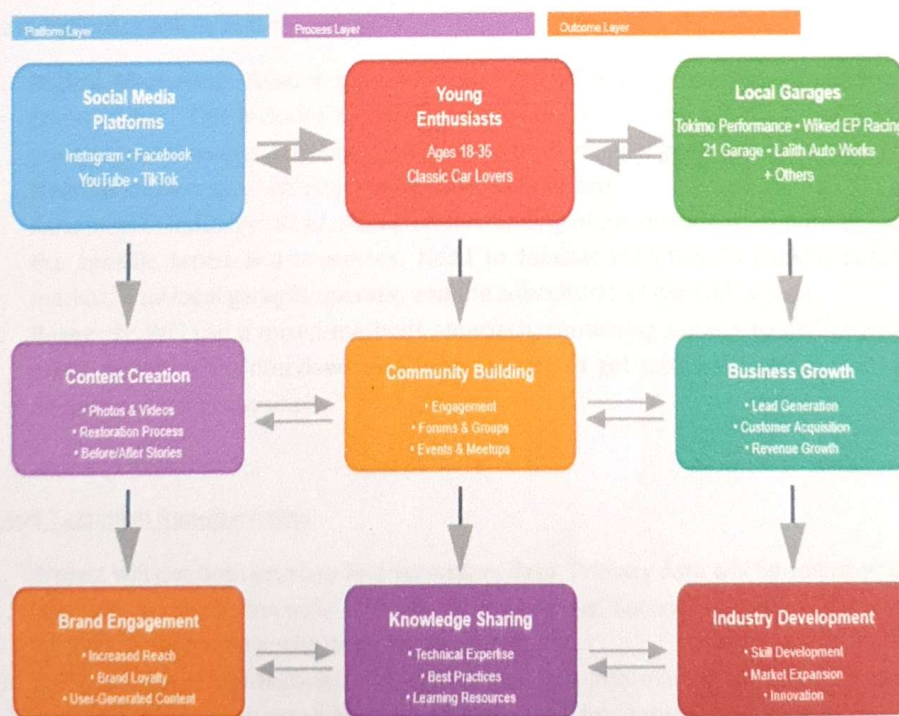
6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed solution involves to develop an effective social media engagement framework that increases young automotive enthusiasts in Sri Lanka's classic and sports car modification and restoration market. This approach will bridge the gap between traditional automotive marketing and community-centric and digitalized emerging marketing engagement.

Solution Includes:

1. Digital culture study: Analysis of existing social media communities and subcultures.
2. Platform- specific engagement techniques: Personalized methods for Instagram, Facebook, YouTube, and TikTok
3. Community-centric marketing approach: Moving from advertising to actively participating in marketing.
4. Sri Lankan garage integration framework: Connecting local restoration and modification workshops by linking online communities.

Social Media Engagement Framework for Sri Lankan Automotive Restoration



Data Flow & Analytics Pipeline

Social Media APIs → Data Processing → Analytics Engine → Business Intelligence → Decision Making

The solution will create a system where social media posts about classic car modification and restoration attract and engage car enthusiasts in Sri Lanka. This community engagement will help local businesses grow and allow young generation to learn and share their expertise.

By understanding the interest and how young car enthusiasts use social media, the framework will enable more effective in the car modification and restoration subculture in Sri Lankan automotive industry.

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

Domain Expertise and Knowledge

The core expertise will lie in three areas:

1. **Digital Marketing:** Need experts who understand how to use social media to build communities. This includes knowing social media algorithms, what content works for young car enthusiasts, and how to use data to track younger generation interest in online. Need to know how to collaborate with local influencers.
2. **Automotive Industry:** Need a deep understanding of car restoration is critical, including the specific terms and processes. Need to familiar with the Sri Lankan automotive market, how local garages operate, and the subcultures of car enthusiasts.
3. **Research:** Will use a mixed-methods approach, combining surveys to gather data from many people with interviews and focus groups to get detailed insights from garage owners and car enthusiasts.

Data and Technical Requirements

1. Project will use both **primary and secondary data**. Primary data will be collected directly from the audience through surveys and interviews. Secondary data will collect from existing social media analytics and industry reports.
2. Will use a variety of **technical tools** to process the information, including social media analytics software, survey platforms, and programs for statistical analysis.
3. Will build a strong **collaborative network** by partnering with local garages and young car enthusiast groups. This ensures the research is relevant and the strategies are practical for the real world.

8. Objectives and Novelty

Main Objective To develop an effective social media engagement framework that increases young automotive enthusiasts in Sri Lanka's classic and sports car modification and restoration market.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Dasuni K.S. IT21385582	To quantify and analyze foundational concepts of social media engagement in automotive restoration communities.	- Literature review on social media engagement theories. - Analysis of existing automotive social media strategies. - Identification of engagement patterns in restoration parties.	Comprehensive study of social media engagement specially for Sri Lankan automotive restoration market.

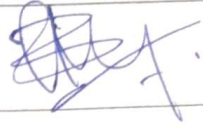
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	To profile the young automotive enthusiasts (ages 18-35) in Sri Lanka and their digital subcultures.	• Survey design and data collection from young enthusiasts. • Digital ethnography of automotive communities. • Behavioral analysis of online interactions.	Novel profiling of Sri Lankan young automotive enthusiasts with focus on digital behavior patterns.
	To analyze the role of major social media platforms in engaging automotive subcultures.	• Platform-specific content analysis. • Engagement metrics evaluation. • Best practices identification for each platform.	Systematic analysis of platform-specific strategies for automotive restoration in developing markets.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Dasuni K.S. IT21385582	System Analysis and Design : This component aligns with (Information Systems Engineering) ISE by conducting comprehensive system analysis of existing social media engagement processes in automobile industry. The research involves requirements gathering, system modeling using UML, Component diagrams, and designing information system architecture for social media integration. Database design theories will be applied to structure engagement data, and design methodologies will be used to develop social media management frameworks for Sri Lankan automobile garages. Focus: System analysis of social media engagement processes
	Data Engineering and Analytics: This component aligns with ISE through the design and implementation of data collection, processing, and analytics systems. This research involves developing ETL (Extract, Transform, Load) processes for social media data, designing data for analytics and implementing machine learning algorithms for behavioral pattern recognition. Focus: Data collection, processing and analytics systems
	Software Engineering and System Integration: This component aligns with ISE by the development of integrated software solutions for social media platform management. The research involves software development lifecycle methodologies, API integration, and creating automated content management systems. Agile development practices will be used to build prototypes for cross-platform connectivity. Focus: Software development and platform integration

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Rangi	Liyanage	
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	☒	No	☐
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- b) Does the proposed topic exhibit novelty?

Yes	☒	No	☐
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes	☒	No	☐
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	☒	No	☐
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

Novelty part need to be improved.

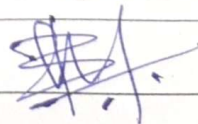
Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	✓
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Need to be improved (Specially, the novelty part) with related to the past papers.

Staff Member's Name	Signature
Rangi Liyanage	

*Important:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.